RANDOM FORESTS

IMPROVING VARIANCE

- Let $X_1, X_2, \ldots, X_n$ be independent random variable each with variance σ^2
- Define $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- What is the variance of $\overline{X}$?

$$Var(\overline{X}) = \frac{\sigma^2}{n}$$

- The using the mean of random variable will result in a lower variance

APPLYING THIS PRINCIPLE TO TREES

- Reduce the variance in the predictions by building a lot trees and averaging the results to produce a final prediction
- But how can we get more data to build a new trees?

- Well maybe we don't have to…

BOOTSTRAP AGGREGATION (ING)

- **Bagging** is an *ensemble** method which uses bootstrap sampling to generate multiple predictions and
    - Take a random sample of the training data *with replacement*
    - Build a tree using the sampled data
    - Make predictions with the model
    - Repeat B times
    - Average predictions across all models for final prediction

**An ensemble method an any method that involves aggregating predictions over multiple models*

MORE ABOUT BAGGING

- Bagging can be using for both regression and classification
  - Regression predictions are the mean
  - Classification predictions are the majority vote
- Bagging can also be used with models that aren’t decision trees

WHAT IF THE PREDICTIONS ***AREN'T*** INDEPENDENT?

- The reduction in variance isn't as strong when variables are correlated

$$Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2Cov(X_1, X_2)$$

- The random forest method is similar to bagging except that it tries to de-correlate the trees through more randomization

ENSEMBLE VARIANCE

- Let $T_1, \ldots, T_M$ be predictions from M tree predictors (for a fixed input x). Denote
  - $\mu = E[T_i]$ (assumed same for all trees),
  - $\sigma^2 = \mathrm{Var}(T_i)$ (same for all trees),
  - $\rho = \mathrm{Corr}(T_i, T_j)$ for $i \neq j$ (average pairwise correlation).
  - $\mathrm{Cov}(T_i, T_j) = \rho\sigma^2$

$$\mathrm{Var}(\bar{T}) = \mathrm{Var}\left(\frac{1}{M}\sum_{i=1}^{M} T_i\right) = \frac{1}{M^2}\mathrm{Var}\left(\sum_{i=1}^{M} T_i\right)$$

$$\boxed{\mathrm{Var}(\bar{T}) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2}$$

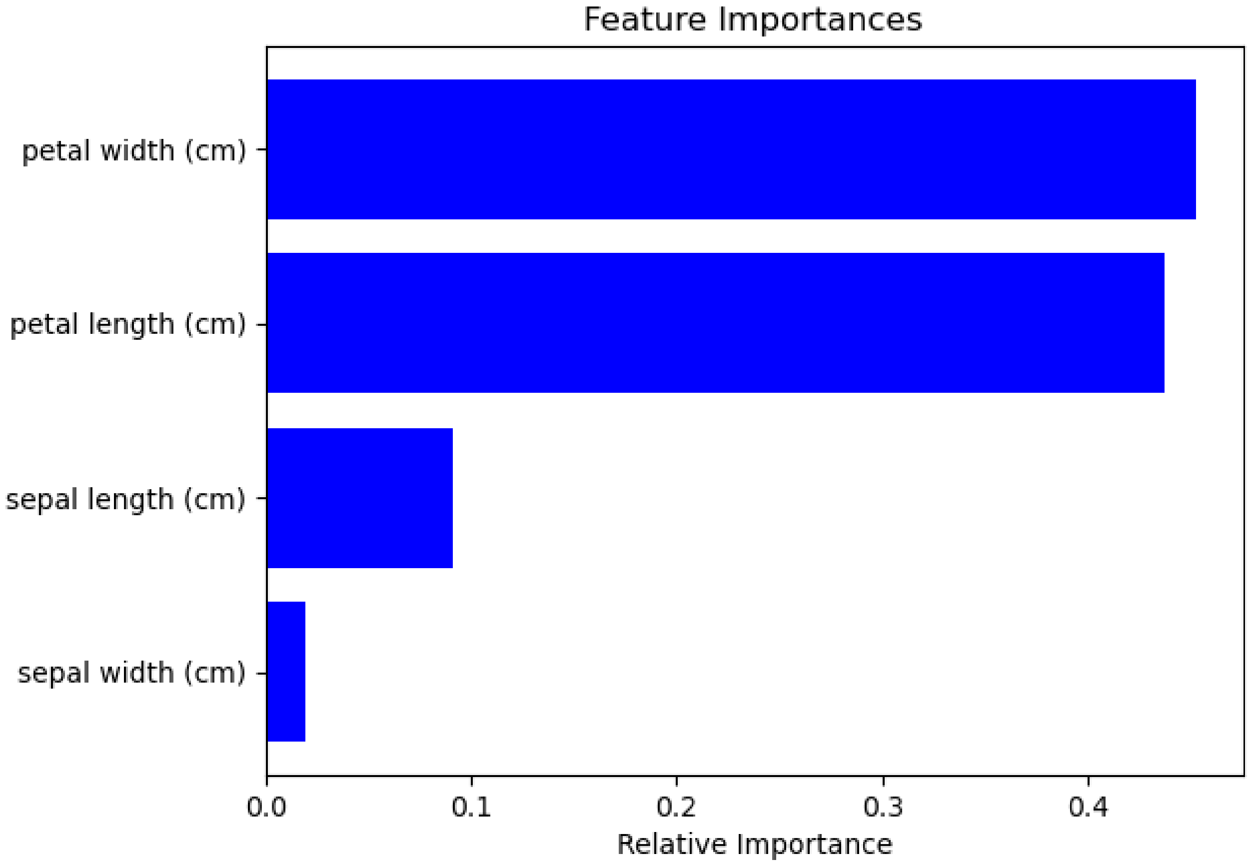
RANDOM FOREST SPLITS

- Take a bootstrap sample of the observations
- Build decision trees, but at every split only consider a **random sample of m features** (from p total features) as split candidates
  - It is typical to use $m \approx \sqrt{p}$
  - Finds alternative paths to decisions
- Rationale:
  - Suppose there a one **very** strong predictor and several moderately strong predictors
  - With bagging, most of the trees will use the very strong feature so most of the predictions will be similar

MODEL INTERPRETATION

- The bad news:
  The increase in predictive ability decreases the interpretability of the model (compared to one tree)
    - We can't just draw the tree anymore
- The good news:
  We can get an overall idea of *feature importance* measured by:
    - Finding splits using a particular feature
    - Computing the average impurity reduction across all trees
    - Features with the largest average decrease in impurity (Gini or entropy) are the most important
    - Scikit-learn scales so that feature importances sum to 1

EXAMPLE OF FEATURE IMPORTANCE



OUT OF BAG ERROR ESTIMATION

- It can be shown that in a bootstrap sample, on average 2/3 of the instances are included
- The remaining 1/3 of the instances are not used to fit a given bagged tree
  - The unused instances are called **out-of-bag** (OOB)
  - We can make predictions instance i using trees in which the i^{th} was OOB
  - A single prediction for instance i is computed through aggregation the trees
  - The overall OOB error is asymptotically ***equivalent*** to performing leave-one-out cross-validation (if the number of trees is large enough)
    - That's like performing cross-validation for free

TUNING PARAMETERS (BAGGING AND RF)

- `n_estimators`: number of trees in forest
  - Increasing does not lead to overfitting
- `criterion`: Gini impurity or entropy
- `max_features`: Number of features to consider at each split
- Tuning parameters associated with base estimator

OTHER VARIATIONS

- Pasting
    - Like bagging, except sample is taken without replacement
- Random patches
    - Like bagging, except a random sample of instances and features is used in training base estimator
- Random subspaces
    - Like bagging, except a random sample of features used is training base estimator
- Extremely random trees
    - Like RF, except use random splits (instead of best threshold)